Third Section.

Transformation of the Theta-Functions.

§ 38. Linear transformations of the function $\eta(\omega)$.

The formulae (3), § 37 lead to the linear transformation of the function $\eta(\omega)$. If

$$(\omega_1', \omega_2') = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\omega_1, \omega_2) \tag{1}$$

is put, then (η_1, η_2) passes into

$$(\eta_1', \eta_2') = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\eta_1, \eta_2) \tag{2}$$

[§ 35, (14)], and $\omega = \omega_2 : \omega_1$ passes into

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}. \tag{3}$$

It now follows from (2), with regard to (3) of the preceding paragraph, that

$$\begin{aligned} \eta_1' &= -\frac{2\pi i}{\omega_1'} \frac{d \log \eta(\omega')}{d\omega'} = -\frac{2\pi i \omega_1'}{\omega_1^2} \frac{d \log \eta(\omega)}{d\omega} - \frac{\pi i \beta}{\omega_1}, \\ \eta_2' &= -\frac{2\pi i \omega_2'}{\omega_1'^2} \frac{d \log \eta(\omega')}{d\omega'} - \frac{\pi i}{\omega_1'} = -\frac{2\pi i \omega_2'}{\omega_1^2} \frac{d \log \eta(\omega)}{d\omega} - \frac{\pi i \delta}{\omega_1}. \end{aligned}$$

These two relations give concordantly

$$\frac{d \log \eta(\omega')}{\omega_1'^2 \, d\omega'} = \frac{d \log \eta(\omega)}{\omega_1^2 \, d\omega} + \frac{\beta}{2\omega_1 \omega_1'},$$

or finally, since by (3)

$$d\omega' = \frac{d\omega}{(\alpha + \beta\omega)^2}, \qquad \omega_1'^2 \, d\omega' = \omega_1^2 \, d\omega,$$

$$\frac{d \log \eta(\omega')}{d\omega} = \frac{d \log \eta(\omega)}{d\omega} + \frac{\beta}{2(\alpha + \beta\omega)}.$$

By integration this gives

$$\eta\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = \varepsilon \sqrt{\alpha + \beta\omega}\, \eta(\omega), \tag{4}$$

where ε is independent of ω, and hence depends only on the numbers $\alpha, \beta, \gamma, \delta$.

The exact determination of this constant ε, particularly with regard to the sign, is a well-known important problem, which presents peculiar difficulties, but whose solution is indispensable for us. Solutions have been given, by various routes, by Hermite,[1] Dedekind,[2] and recently also by Mertens and Scheibner.[3] We shall here take a route which has the advantage of great simplicity, though admittedly it contains not a derivation, but only a proof of the completed formula.

[1] Liouville's Journal, Ser. II, vol. III, 1858; *Oeuvres de Charles Hermite*, p. 487.

[2] Explanations to no. XXVIII of Riemann's works, second edition, and "Über die elliptischen Modulfunktionen", Crelle's Journal, vol. 83, p. 265. Compare also the author's paper "Zur Theorie der elliptischen Funktionen", Acta Mathematica, vol. 6, pp. 341 ff.

[3] Mertens, "Zur linearen Transformation der ϑ-Reihen", Transactions of the American Mathematical Society, July 1901. Scheibner, "Zur linearen Transformation der Theta-Funktionen und elliptischen Modulfunktionen", Berichte der Sächsischen Gesellschaft der Wissenschaften, October 1906.

For two special cases we have already carried out this determination earlier [§ 34, (4), (5)], and we shall base ourselves here on that result. The formulae are

$$\eta(\omega \pm 1) = e^{\pm\pi i/12}\eta(\omega) \tag{5}$$

and

$$\eta\left(-\frac{1}{\omega}\right) = \sqrt{-i\omega}\,\eta(\omega), \tag{6}$$

where $\sqrt{-i\omega}$ is to be taken with positive real part.

We now put

$$\frac{\eta\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right)}{\eta(\omega)} = E\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}; \omega\Big), \tag{7}$$

and have to determine the value of this symbol. By (5), (6), (7), one has

$$E\begin{pmatrix}-\alpha & -\beta\\ -\gamma & -\delta\end{pmatrix}; \omega\Big) = E\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}; \omega\Big), \tag{8}$$

$$E\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}; \omega\Big) = 1, \tag{9}$$

$$E\begin{pmatrix}1 & 0\\ \pm 1 & 1\end{pmatrix}; \omega\Big) = e^{\pm\pi i/12}, \tag{10}$$

$$E\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}; \omega\Big) = \sqrt{-i\omega}. \tag{11}$$

We now consider two substitutions and their product:

$$\begin{pmatrix}\alpha'' & \beta''\\ \gamma'' & \delta''\end{pmatrix} = \begin{pmatrix}\alpha' & \beta'\\ \gamma' & \delta'\end{pmatrix}\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}.$$

If

$$\omega' = \frac{\gamma+\delta\omega}{\alpha+\beta\omega},$$

then from the definition (7) one obtains

$$E\begin{pmatrix}\alpha'' & \beta''\\ \gamma'' & \delta''\end{pmatrix}; \omega\Big) = E\begin{pmatrix}\alpha' & \beta'\\ \gamma' & \delta'\end{pmatrix}; \omega'\Big)\, E\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}; \omega\Big). \tag{12}$$

Two special cases are obtained from this by putting

$$\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix} = \begin{pmatrix}1 & 0\\ \pm 1 & 1\end{pmatrix}, \begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix},$$

and then writing again $\alpha, \beta, \gamma, \delta$ in place of $\alpha', \beta', \gamma', \delta'$. Thus

$$E\begin{pmatrix}\alpha \pm \beta & \beta\\ \gamma \pm \delta & \delta\end{pmatrix}; \omega\Big) = e^{\pm\pi i/12} E\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}; \omega \pm 1\Big), \tag{13}$$

$$E\begin{pmatrix}-\beta & \alpha\\ -\delta & \gamma\end{pmatrix}; \omega\Big) = \sqrt{-i\omega}\, E\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}; -\frac{1}{\omega}\Big). \tag{14}$$

It was shown in § 30 that all linear transformations can be composed by repeated application of the two fundamental transformations

$$\begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}, \qquad \begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}$$

and of their inverse transformations. Therefore, by (12), the formulae (8) to (14) completely define the symbol E.

Thus, if we know an expression satisfying these conditions, it must agree with E. In order to set up such an expression, we distinguish two cases. Since α, β are relatively prime, one of them is certainly odd. We put:

1. α odd and positive:
$$E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; \omega\Big) = \left(\frac{\beta}{\alpha}\right) i^{(\alpha-1)/2} e^{\frac{\pi i}{12}[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]}\sqrt{\alpha+\beta\omega},$$

2. β odd and positive:
$$E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; \omega\Big) = \left(\frac{\alpha}{\beta}\right) i^{(1-\beta)/2} e^{\frac{\pi i}{12}[\beta(\alpha+\delta)-(\beta^2-1)\alpha\gamma]}\sqrt{-i(\alpha+\beta\omega)}. \tag{15}$$

Here the following is still to be noted. The roots $\sqrt{\alpha+\beta\omega}$ and $\sqrt{-i(\alpha+\beta\omega)}$ are to be taken with positive real part. That one of them could be purely imaginary is excluded by the assumption that ω has positive imaginary part and that α, respectively β, is positive; for then $\alpha+\beta\omega$, respectively $-i(\alpha+\beta\omega)$, cannot be real and negative. The signs $\left(\frac{\beta}{\alpha}\right)$ and $\left(\frac{\alpha}{\beta}\right)$ denote the Legendre-Jacobi symbol from the theory of quadratic residues, with the extension that $\left(\frac{\beta}{1}\right)$ and $\left(\frac{0}{1}\right)$ are to be equal to 1. If in the first case α, or in the second β, is negative, then on the right all signs of $\alpha, \beta, \gamma, \delta$ are to be changed. If both α and β are odd, then both (15), 1 and (15), 2 can be applied, and both give, as one easily verifies by means of the reciprocity law for quadratic residues, the same result.

It is namely true, when α and β are odd, α is assumed positive, and the upper or lower sign is to be taken according as β is positive or negative, that

$$\left(\frac{\beta}{\alpha}\right)\left(\frac{\pm\alpha}{\pm\beta}\right) = \pm e^{-\pi i(\alpha+1)(\beta-1)/4},$$

$$\sqrt{\pm i(\alpha+\beta\omega)} = e^{\pm\pi i/4}\sqrt{\alpha+\beta\omega},$$

and the identity of (15), 1 and 2 then follows from the congruences

$$-3\alpha\beta+\beta(\alpha+\delta)-(\beta^2-1)\alpha\gamma \equiv \alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta \pmod{24},$$
$$\alpha\gamma+\alpha\beta \equiv \beta\delta \pmod 8,$$

of which the second, if α and β are both not divisible by 3, and hence $\alpha^2 \equiv \beta^2 \equiv 1 \pmod 3$, is also valid for the modulus 24.

That (15) satisfies the formulae (8) to (11) is seen immediately; it remains to show that (13) and (14) are fulfilled. We begin with (14), where it may be assumed that α is odd and positive. If in (14) one interchanges ω with $-1/\omega$, α and $-\beta$ are interchanged, and these cannot both be even. It is

$$\sqrt{-i\omega}\sqrt{\alpha-\frac{\beta}{\omega}} = \sqrt{-i(-\beta+\alpha\omega)};$$

indeed, putting temporarily

$$-i\omega = re^{i\varphi}, \qquad \alpha-\frac{\beta}{\omega} = \rho e^{i\psi}, \qquad -i(-\beta+\alpha\omega) = r\rho e^{i(\varphi+\psi)},$$

the real parts of $-i\omega$ and $-i(-\beta+\alpha\omega)$ being positive imply

$$-\frac{\pi}{2} < \varphi < \frac{\pi}{2}, \quad -\pi < \psi < \pi, \quad -\frac{\pi}{2} < \varphi+\psi < \frac{\pi}{2},$$

so that

$$\sqrt{-i\omega} = \sqrt{r}e^{i\varphi/2}, \qquad \sqrt{\alpha-\frac{\beta}{\omega}} = \sqrt{\rho}e^{i\psi/2},$$

and the real part of their product is positive. Consequently (15), 1 gives

$$\sqrt{-i\omega}E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; -\frac{1}{\omega}\Bigg) = \left(\frac{\beta}{\alpha}\right) i^{(\alpha-1)/2} e^{\frac{\pi i}{12}[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]} \times \sqrt{-i(-\beta+\alpha\omega)}.$$

The same formula is obtained, since

$$\left(\frac{-\beta}{\alpha}\right) = i^{\alpha-1}\left(\frac{\beta}{\alpha}\right),$$

from (15), 2 for

$$E\begin{pmatrix}-\beta & \alpha \\ -\delta & \gamma\end{pmatrix}; \omega\Bigg),$$

and so (14) is proved.

There remains formula (13). It is enough to consider only the upper signs, that is, to prove

$$E\begin{pmatrix}\alpha+\beta & \beta \\ \gamma+\delta & \delta\end{pmatrix}; \omega\Bigg) = e^{\pi i/12} E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; \omega+1\Bigg),$$

for the other case is reduced to this one by replacing α, γ, ω with $\alpha+\beta, \gamma+\delta, \omega+1$. If first β is odd and positive, then (13) follows from (15), 2 by the congruence

$$(\beta^2-1)(1-\beta\gamma-\alpha\delta-\beta\delta) \equiv -\beta(\beta^2-1)(2\gamma+\delta) \equiv 0 \pmod{24}.$$

If β is even, then α is odd. Suppose α positive; then $\alpha+\beta$ may be positive or negative. If in the first case the upper signs, in the second case the lower signs are valid, (15), 1 gives

$$e^{\pi i/12} E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; \omega+1\Bigg) = \left(\frac{\beta}{\alpha}\right) i^{(1-\alpha)/2} e^{\frac{\pi i}{12}[1+\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]}\sqrt{\alpha+\beta+\beta\omega}, \tag{16}$$

whereas

$$E\begin{pmatrix}\alpha+\beta & \beta \\ \gamma+\delta & \delta\end{pmatrix}; \omega\Bigg) = \left(\frac{\beta}{\pm(\alpha+\beta)}\right) i^{\{1\mp(\alpha+\beta)\}/2} e^{\frac{\pi i}{12}\{(\alpha+\beta)(\gamma+\delta-\beta)-[(\alpha+\beta)^2-1]\beta\delta\}} \times \sqrt{\pm(\alpha+\beta+\beta\omega)}. \tag{17}$$

If the lower signs hold, β is negative, and on putting

$$-(\alpha+\beta+\beta\omega) = re^{i\varphi}, \qquad i(\alpha+\beta+\beta\omega) = re^{i(\varphi-\pi/2)},$$

we have to set

$$\sqrt{-(\alpha+\beta+\beta\omega)} = i\sqrt{\alpha+\beta+\beta\omega}.$$

Furthermore, by the reciprocity law for quadratic residues,[4]

$$\left(\frac{\beta}{\alpha+\beta}\right) = (-1)^{\beta(\alpha+1)/4}\left(\frac{\beta}{\alpha}\right),$$

[4]By the reciprocity law, if $\beta = \pm 2^\lambda \beta'$ is put and β' is assumed odd and positive, then, when $\alpha+\beta$ is positive,

$$\left(\frac{\beta}{\alpha+\beta}\right) = \left(\frac{\pm 2^\lambda}{\alpha+\beta}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{(\alpha+\beta-1)(\beta'-1)}{4}}, \qquad \left(\frac{\beta}{\alpha}\right) = \left(\frac{\pm 2^\lambda}{\alpha}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{(\alpha-1)(\beta'-1)}{4}}.$$

If $\lambda \geq 2$, these two values are equal; if $\lambda = 1$, they differ by the factor $(-1)^{(\alpha+1)/2}$, in agreement with the first of the preceding formulae. The correctness of the second formula, when $\alpha+\beta$ is negative, follows from

$$\left(\frac{-\beta}{-(\alpha+\beta)}\right) = \left(\frac{-2^\lambda}{-(\alpha+\beta)}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{(\alpha+\beta-1)(\beta'-1)}{4}}, \qquad \left(\frac{\beta}{\alpha}\right) = \left(\frac{2^\lambda}{\alpha}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{\alpha-1}{2}}(-1)^{\frac{(\alpha-1)(\beta'-1)}{4}}.$$

$$\left(\frac{-\beta}{-(\alpha+\beta)}\right) = (-1)^{(\alpha-1)/2}(-1)^{\beta(\alpha-1)/4}\left(\frac{\beta}{\alpha}\right).$$

It follows that the two expressions (16), (17) agree, and consequently (13) is correct, by the congruence

$$\beta(3\alpha - 2\gamma + \beta - \delta + \beta^2\delta + 2\alpha\beta\delta) \equiv 0 \pmod{24},$$

which, since β is assumed even, follows from $\alpha\delta - \beta\gamma = 1$. Thus the formulae (15) are proved.

If we put

$$E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; \omega) = \varepsilon\sqrt{\alpha + \beta\omega}, \tag{18}$$

then ε is a twenty-fourth root of unity, whose product with $\sqrt{\alpha+\beta\omega}$ is completely determined by (15), and the transformation of the η-function is

$$\eta\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \varepsilon\sqrt{\alpha+\beta\omega}\,\eta(\omega). \tag{19}$$

For ε^{12} one finds

$$\varepsilon^{12} = (-1)^{\alpha\beta+\gamma\delta+\beta\gamma}. \tag{20}$$

§ 39. Linear transformation of the ϑ-functions.

The transformation formulae of the ϑ-functions are consequences of the fundamental properties of the σ-function: the latter remains unchanged under the substitution

$$(\omega_1', \omega_2') = \begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}(\omega_1, \omega_2).$$

From § 37, (4), replacing $\omega_1, \omega_2, \eta_1$ by $\omega_1', \omega_2', \eta_1'$, one obtains

$$\frac{\omega_1 e^{\eta_1 u^2/\omega_1}\vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{11}'\left(0, \frac{\omega_2}{\omega_1}\right)} = \frac{\omega_1' e^{\eta_1' u^2/\omega_1'}\vartheta_{11}\left(\frac{u}{\omega_1'}, \frac{\omega_2'}{\omega_1'}\right)}{\vartheta_{11}'\left(0, \frac{\omega_2'}{\omega_1'}\right)}.$$

Since, by § 35, (10), (13), (14),

$$\frac{\eta_1}{\omega_1} - \frac{\eta_1'}{\omega_1'} = \frac{\omega_1'\eta_1 - \eta_1'\omega_1}{\omega_1\omega_1'} = \beta\frac{\omega_2\eta_1 - \omega_1\eta_2}{\omega_1\omega_1'} = \frac{\pi i\beta}{\omega_1\omega_1'}, \tag{1}$$

if we put

$$v = \frac{u}{\omega_1}, \qquad \omega = \frac{\omega_2}{\omega_1}, \qquad v' = \frac{u}{\omega_1'} = \frac{v}{\alpha+\beta\omega}, \qquad \omega' = \frac{\omega_2'}{\omega_1'} = \frac{\gamma+\delta\omega}{\alpha+\beta\omega}, \tag{2}$$

then

$$\frac{\vartheta_{11}(v', \omega')}{\vartheta_{11}'(0, \omega')} = \frac{e^{\pi i\beta v v'}}{\alpha+\beta\omega}\frac{\vartheta_{11}(v, \omega)}{\vartheta_{11}'(0, \omega)}.$$

Moreover

$$\vartheta_{11}'(0, \omega) = 2\pi\eta(\omega)^3, \qquad \vartheta_{11}'(0, \omega') = 2\pi\eta(\omega')^3,$$

and hence by § 38, (19)

$$\vartheta_{11}'(0, \omega') = \varepsilon^3(\alpha+\beta\omega)^{3/2}\vartheta_{11}'(0, \omega).$$

Thus finally

$$e^{-\pi i\beta v v'}\vartheta_{11}(v', \omega') = \varepsilon^3(\alpha+\beta\omega)^{1/2}\vartheta_{11}(v, \omega). \tag{3}$$

The transformation formulae for the other three functions $\vartheta_{00}, \vartheta_{01}, \vartheta_{10}$ differ in the six classes of § 36 and can be derived from formulae (5), § 36 in the same way. But the six cases can also be assembled into a single formula-system by replacing v in (3) by

$$v + \frac{\alpha + \beta\omega}{2}, \qquad v + \frac{\gamma + \delta\omega}{2}, \qquad v + \frac{\alpha + \gamma + (\beta + \delta)\omega}{2},$$

and hence replacing v' respectively by

$$v' + \frac{1}{2}, \qquad v' + \frac{\omega'}{2}, \qquad v' + \frac{1 + \omega'}{2},$$

and then applying the formulae (2), (3), (8) of § 21 on both the right and left sides of (3). This gives

$$e^{-\pi i\beta vv'}\vartheta_{10}(v', \omega') = i^{\delta} e^{-\pi i\alpha\beta/4}\varepsilon^3\sqrt{\alpha + \beta\omega}\,\vartheta_{1+\delta,1-\alpha}(v, \omega), \tag{4}$$

$$e^{-\pi i\beta vv'}\vartheta_{01}(v', \omega') = i^{\delta-1} e^{-\pi i\gamma\delta/4}\varepsilon^3\sqrt{\alpha + \beta\omega}\,\vartheta_{1+\delta,1-\gamma}(v, \omega), \tag{5}$$

$$e^{-\pi i\beta vv'}\vartheta_{00}(v', \omega') = i^{\delta+\beta-\alpha\delta} e^{-\pi i(\alpha\beta+\gamma\delta)/4}\varepsilon^3\sqrt{\alpha + \beta\omega}\,\vartheta_{1+\beta+\delta,1-\alpha-\gamma}(v, \omega). \tag{6}$$

The fourth powers of these functions are expressible more simply:

$$e^{-4\pi i\beta vv'}\vartheta^4_{g_1,g_2}(v', \omega') = (-1)^{g_2\alpha\beta+g_1\gamma\delta+\beta\gamma}(\alpha + \beta\omega)^2\vartheta^4_{g'_1,g'_2}(v, \omega), \tag{7}$$

where, in the six classes, the characteristics (g'_1, g'_2) belonging to $(g_1, g_2) = (00), (01), (10)$ are to be taken from the last column of the table in § 36.

A simpler transformation formula is obtained from the σ-functions for the product of the three ϑ-functions (4), (5), (6). In fact, forming the product of the three functions § 37, (6), one has

$$\sigma_{00}(u)\sigma_{01}(u)\sigma_{10}(u) = e^{3\eta_1 u^2/\omega_1}\frac{\vartheta_{00}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)\vartheta_{01}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)\vartheta_{10}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{00}\vartheta_{01}\vartheta_{10}},$$

a function which, by § 36, remains completely unchanged under linear transformation. Using in addition the relation

$$\eta'_1\omega_1 - \eta_1\omega'_1 = -\pi i\beta,$$

one obtains

$$\frac{\vartheta_{00}(v, \omega)\vartheta_{01}(v, \omega)\vartheta_{10}(v, \omega)}{\vartheta_{00}\vartheta_{01}\vartheta_{10}} = e^{-3\pi i\beta vv'}\frac{\vartheta_{00}(v', \omega')\vartheta_{01}(v', \omega')\vartheta_{10}(v', \omega')}{\vartheta_{00}(0, \omega')\vartheta_{01}(0, \omega')\vartheta_{10}(0, \omega')}. \tag{8}$$

Here one may still put, by (19) of the preceding paragraph,

$$\vartheta_{00}(0, \omega')\vartheta_{01}(0, \omega')\vartheta_{10}(0, \omega') = \varepsilon^3(\alpha + \beta\omega)^{3/2}\vartheta_{00}\vartheta_{01}\vartheta_{10}.$$

But from (8) we draw another conclusion. If one puts

$$v' = \frac{h}{n}, \qquad v = \frac{h(\alpha + \beta\omega)}{n},$$

and takes the product for $h = 1, 2, \ldots, (n-1)/2$, then, using the known relation

$$\sum_{h=1}^{(n-1)/2} h^2 = \frac{n(n^2 - 1)}{24}$$

and applying on the right-hand side of (8) the formula (23), § 32, one obtains

$$2^{(n-1)/2}\prod_{h=1}^{(n-1)/2}\vartheta_{00}\left(\frac{h(\alpha + \beta\omega)}{n}\right)\vartheta_{01}\left(\frac{h(\alpha + \beta\omega)}{n}\right)\vartheta_{10}\left(\frac{h(\alpha + \beta\omega)}{n}\right) = e^{-\frac{\pi i}{8}\frac{n^2-1}{n}\beta(\alpha+\beta\omega)}\vartheta_{00}^{(n-1)/2}\vartheta_{10}^{(n-1)/2}\vartheta_{01}^{(n-1)/2} \tag{9}$$

where now α, β may be any relatively prime pair.

§ 40. Linear transformation of the functions $f(\omega), f_1(\omega), f_2(\omega)$.

By § 34, (9), the formulae for the linear transformation of the f-functions are a simple consequence of the transformation of the η-function. Let us first suppose, in the linear transformation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix},$$

that β is even, and therefore α, δ are odd. Then

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} \alpha & \frac{1}{2}\beta \\ 2\gamma & \delta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \tag{1}$$

that is,

$$2\frac{\gamma + \delta\omega}{\alpha + \beta\omega} = \frac{2\gamma + \delta \cdot 2\omega}{\alpha + \frac{1}{2}\beta \cdot 2\omega}.$$

But by § 34, (9),

$$f_2(\omega) = \sqrt[6]{2}\frac{\eta(2\omega)}{\eta(\omega)}.$$

If therefore the substitution $\omega, (\gamma+\delta\omega)/(\alpha+\beta\omega)$ is made, then with the notation of § 38 one obtains

$$f_2\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = \frac{E\begin{pmatrix} \alpha & \frac{1}{2}\beta \\ 2\gamma & \delta \end{pmatrix}; 2\omega\end{pmatrix}}{E\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}; \omega\end{pmatrix}} f_2(\omega). \tag{2}$$

Here the E-functions are to be determined by § 38, (15), 1, from which

$$\frac{E\begin{pmatrix} \alpha & \frac{1}{2}\beta \\ 2\gamma & \delta \end{pmatrix}; 2\omega\end{pmatrix}}{E\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}; \omega\end{pmatrix}} = \left(\frac{2}{\alpha}\right) e^{\frac{\pi i}{12}\left(\alpha\gamma + \frac{\alpha\beta}{2} + \frac{(\alpha^2-1)\beta\delta}{2}\right)}.$$

But $\frac{1}{4} = \frac{3}{8} - \frac{1}{3}$, and

$$\begin{aligned} \alpha\gamma + \frac{\alpha\beta}{2} + \frac{(\alpha^2-1)\beta\delta}{2} &\equiv \frac{\alpha(2\gamma+\beta)}{2} \pmod 8, \\ &\equiv \alpha(\gamma - \beta) - (\alpha^2 - 1)\beta\delta \pmod 3. \end{aligned}$$

Thus, putting for abbreviation

$$\rho = e^{\frac{2\pi i}{3}[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]}, \tag{3}$$

we get from (2)

$$f_2\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = \left(\frac{2}{\alpha}\right) \rho e^{\frac{3\pi i}{8}\alpha(2\gamma+\beta)} f_2(\omega), \qquad \beta \equiv 0 \pmod 2. \tag{4}$$

In the same way all other formulae of this kind can be derived. They are obtained more simply from (4) itself with the aid of the fundamental transformations § 34, (13), (14), (15). Thus, if in (4) one replaces ω by $-1 : \omega$ and then interchanges $\alpha, \beta, \gamma, \delta$ with $\beta, -\alpha, \delta, -\gamma$ (whereby ρ remains unchanged), one obtains

$$f_2\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = \left(\frac{2}{\beta}\right) \rho e^{\frac{3\pi i}{8}\beta(2\delta-\alpha)} f_1(\omega), \qquad \alpha \equiv 0 \pmod 2, \tag{5}$$

and if in this ω is replaced by $\omega+3$ and γ, α by $\gamma-3\delta, \alpha-3\beta$, then

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = -\rho e^{\frac{3\pi i}{8}\beta(2\delta-\alpha)} f(\omega), \qquad \alpha-\beta\equiv 0 \pmod 2. \tag{6}$$

If in (4), (5), (6) one puts

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = f_1\left(-\frac{\alpha+\beta\omega}{\gamma+\delta\omega}\right),$$

and then interchanges $\alpha, \beta, \gamma, \delta$ with $-\gamma, -\delta, \alpha, \beta$, one obtains

$$f_1\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \left(\frac{2}{\gamma}\right)\rho e^{-\frac{3\pi i}{8}\gamma(2\alpha-\delta)} f_2(\omega), \qquad \delta\equiv 0 \pmod 2, \tag{7}$$

$$= \left(\frac{2}{\delta}\right)\rho e^{-\frac{3\pi i}{8}\delta(2\beta+\gamma)} f_1(\omega), \qquad \gamma\equiv 0 \pmod 2, \tag{8}$$

$$= -\rho e^{-\frac{3\pi i}{8}\delta(2\beta+\gamma)} f(\omega), \qquad \gamma-\delta\equiv 0 \pmod 2. \tag{9}$$

From (9) and (6), replacing ω by

$$\frac{-\gamma+\alpha\omega}{\delta-\beta\omega}$$

and then $\alpha, \beta, \gamma, \delta$ by $\delta, -\beta, -\gamma, \alpha$, one obtains

$$f\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = -\rho e^{-\frac{3\pi i}{8}\alpha(2\beta+\gamma)} f_1(\omega), \qquad \alpha-\gamma\equiv 0 \pmod 2, \tag{10}$$

$$f\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = -\rho e^{\frac{3\pi i}{8}\beta(2\alpha-\delta)} f_2(\omega), \qquad \beta-\delta\equiv 0 \pmod 2. \tag{11}$$

Finally, if in (10) ω is replaced by $\omega+9$ and γ, α by $\gamma-9\delta, \alpha-9\beta$, then

$$f\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \rho\left(\frac{2}{\alpha-\beta}\right) e^{-\frac{3\pi i}{8}(\alpha-\beta)(\alpha+\beta+\gamma-\delta)} f(\omega), \qquad \alpha+\beta+\gamma-\delta\equiv 0 \pmod 2. \tag{12}$$

The functions introduced by Hermite, $\varphi(\omega), \psi(\omega), \chi(\omega)$, are connected with $f(\omega), f_1(\omega), f_2(\omega)$ by the equations

$$f(\omega) = \frac{\sqrt[6]{2}}{\chi(\omega)}, \qquad f_1(\omega) = \sqrt[6]{2}\frac{\psi(\omega)}{\chi(\omega)}, \qquad f_2(\omega) = \sqrt[6]{2}\frac{\varphi(\omega)}{\chi(\omega)}.$$

The transformation formulae of the f-functions can be reshaped in many ways with use of the relation $\alpha\delta-\beta\gamma=1$. Thus the formulae given by Hermite are not immediately recognizable as identical with ours; a simple calculation, however, shows their agreement. The formulae given above have the advantage that, without losing simplicity, they combine two of the six transformation classes at a time in one expression.